

GLUCOSE — THE SUGAR THAT POWERS LIFE

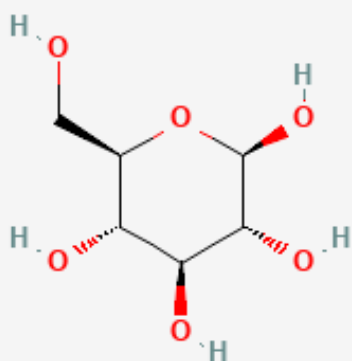
COMPOUND IDENTITY

Common name: Glucose / D-Glucose
IUPAC name: (3R,4S,5S,6R)-6-(hydroxymethyl)oxane-2,3,4,5-tetrol
Molecular formula: $C_6H_{12}O_6$
Molar mass: 180.16 g/mol
CAS number: 50-99-7

IMPORTANT PROPERTIES

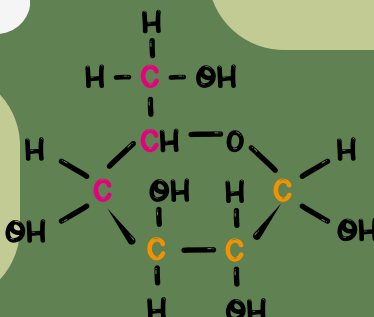
- White/colorless crystalline solid in its pure form
- Highly soluble in water
- Polar
- Molecular formula: $C_6H_{12}O_6$
- Molar mass: 180.16 g/mol

STRUCTURE



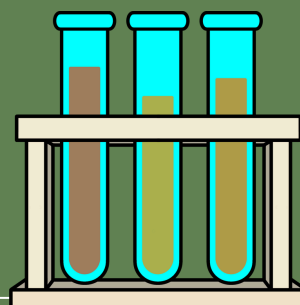
Functional groups/features:

- Multiple hydroxyl ($-OH$) groups
- Ether oxygen in the ring
- Six-carbon carbohydrate structure



USES & VALUE

- Energy source in biological systems
- Food and beverage production
- Pharmaceutical formulations
- Fermentation feedstock
- Production of other biochemical products



WHAT IS AMAZING ABOUT IT?

Glucose is a major chemical fuel for living cells.

Glucose can undergo metabolic reactions that release energy for biological processes.

GLUCOSE ➡ Metabolic Reactions ➡ **ENERGY**

⚠️ RISKS & SAFETY

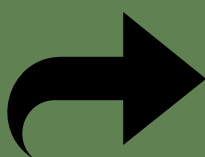
Generally handled as a common carbohydrate, but concentrated glucose products and laboratory chemicals should still be handled appropriately

CHEMICAL ENGINEERING CONNECTION

Fermentation & Bioprocessing

GLUCOSE ➡ **FERMENTATION** ➡ **DESIRED PRODUCTS**

Glucose can serve as a feedstock in fermentation processes. Chemical engineers can work with:



- Bioreactors
- Fermentation
- Separation
- Purification
- Process control

THE SAME MOLECULE THAT GIVES CELLS CHEMICAL ENERGY CAN ALSO SERVE AS A FEEDSTOCK FOR INDUSTRIAL BIOPROCESSES

REFERENCES

Informatics, N. O. O. D. A. (n.d.-b). Benzene. <https://webbook.nist.gov/cgi/cbook.cgi?ID=71-43-2>
PubChem. (n.d.). Benzene. PubChem. <https://pubchem.ncbi.nlm.nih.gov/compound/241>